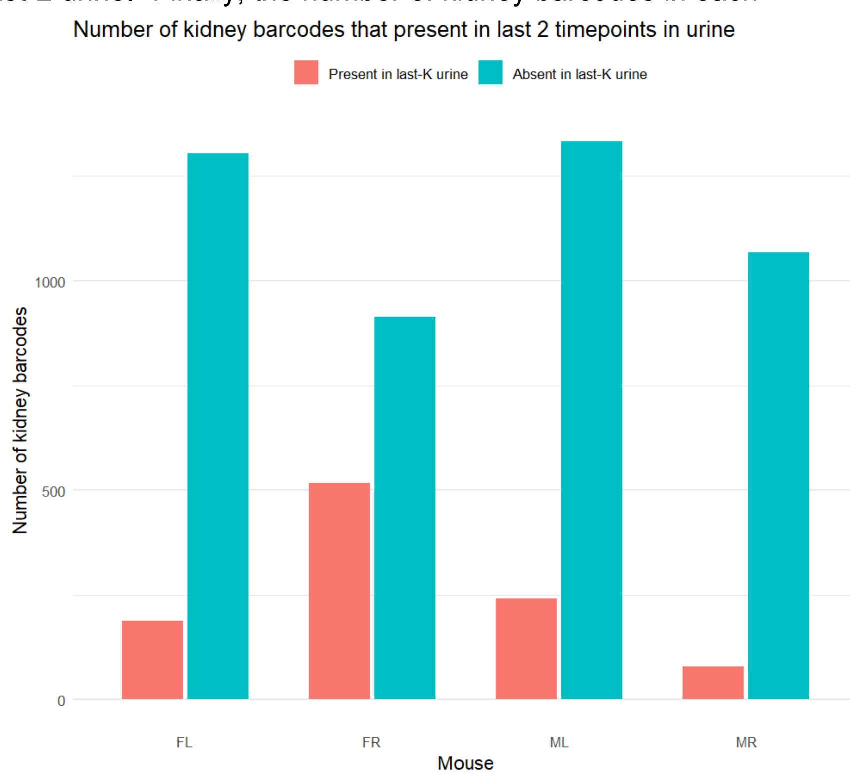
Comprehensive documentation of Kimin Wu Nguyen's analysis and independent validation of results

Kimin Wu Nguyen

Figure 1A

Purpose: Determining in each animal (female left, female right, male left, male right) whether barcodes detected in kidney tissue are also detectable in urine during the last two urine collection timepoints. By restricting the urine window to the last K= 2, the analysis specifically tests whether kidney-resident barcodes are reflected in the most recent urinary output, allowing a per-animal comparison of concordance between kidney tissue and late-stage urine shedding.

Procedure: For each animal independently, the code first identifies the last two post-infection timepoints available in the urine dataset by ordering `Days_pi` and selecting the final two timepoints. Using these timepoints, it constructs the set of barcodes present in kidney tissue by filtering the tissue dataset to kidney samples and retaining barcodes with positive abundance in the last 2 timepoints and summarized per animal. The kidney barcode set is then compared against the last-two-timepoint urine barcode set for the same animal: kidney barcodes that appear in urine during this window are labeled as "Present in last-2 urine," while those that do not are labeled as "Absent in last-2 urine." Finally, the number of kidney barcodes in each category is counted per animal (FL, FR, ML, MR) and visualized as a side-by-side bar plot to facilitate comparison across animals and between presence versus absence in late urine samples.



Result: By using the last two urine timepoints for each animal, the majority of barcodes detected in kidney tissue are absent from urine across all four animals (female left, female right, male left, male right). Only a small fraction of kidney barcodes are detected in urine at these final timepoints.

Among the animals, female right (FR) shows the highest number of kidney barcodes that are also present in late urine samples, while male right (MR) shows the lowest. Overall, kidney–urine barcode overlap at the final two urine collections is limited and varies substantially between animals.

Additional Analysis

Purpose: Determine if urine barcodes detected in the last two timepoints are shared across animals or unique to each animal.

Procedure: Group the barcodes that are presented in last two timepoints across each animal. Then, the barcode sets are compared using a Venn diagram to visualize shared and unique barcodes across animals.

Result: Most urine barcodes detected in the last two timepoints are unique to individual animals, as indicated by the large non-overlapping regions. However, there is a small set of barcodes (3 variants) present in all four animals, showing that while late-stage urine barcodes are largely animal-specific, a limited number are shared across all animals.

	barcode	n_animals
	<chr>	<int>
1	ACATATTGGAAG	4
2	GTACATGATTAG	4
3	TGTTT	4

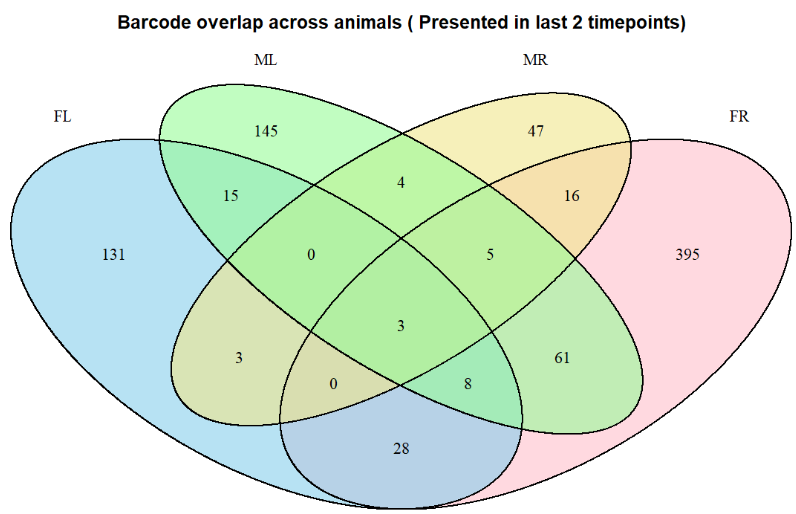


Figure 2A

Purpose: For each animal, this code identifies the top 50 barcodes ranked by kidney abundance that were detectable in urine during the final two days of the study and visualizes how those kidney-dominant barcodes appear in urine over time.

Procedure: For each animal, barcodes detected in urine during the final two post-infection days were identified and used to filter kidney barcode data. Kidney barcodes were then summed across tissue samples, ranked by total abundance, and the top 50 barcodes that were also present in late-stage urine were selected. The abundance of these kidney-enriched barcodes was subsequently tracked in urine across all time points using donut plots, with highlighted segments indicating the selected barcodes and background segments representing all remaining urine barcodes.

Result: Based on the analysis of urinary barcodes, the results show that the barcodes dominating the urine population at late stages of infection generally do not exhibit strong shedding during the early post-infection period. As shown in **Figure 2A**, the urine populations for all four animals (FL, FR, ML, and MR) are initially characterized by a high diversity of "background" barcodes, represented by the fragmented gray segments in the early-stage donuts . During the first 12 days post-infection, the 50 barcodes pre-selected for their high kidney abundance and late-stage urinary presence are either undetectable or present at extremely low frequencies . However, as the infection progresses, a distinct shift in clonal dominance occurs. In animal **MR**, for example, specific kidney-enriched barcodes (indicated by bright blue and orange segments) begin to expand significantly after Day 21, eventually displacing the initial background population to account for nearly the entire urinary output by Day 59 . Similar patterns of delayed expansion are observed in **ML**, where the colored kidney barcodes only achieve substantial prevalence after Day 81 . These findings suggest that the barcodes most successful in colonizing the kidney and persisting into late-stage urine are not necessarily the most "fit" for early shedding; rather, they appear to undergo a period of gradual expansion or selective enrichment before reaching high prevalence in the urine

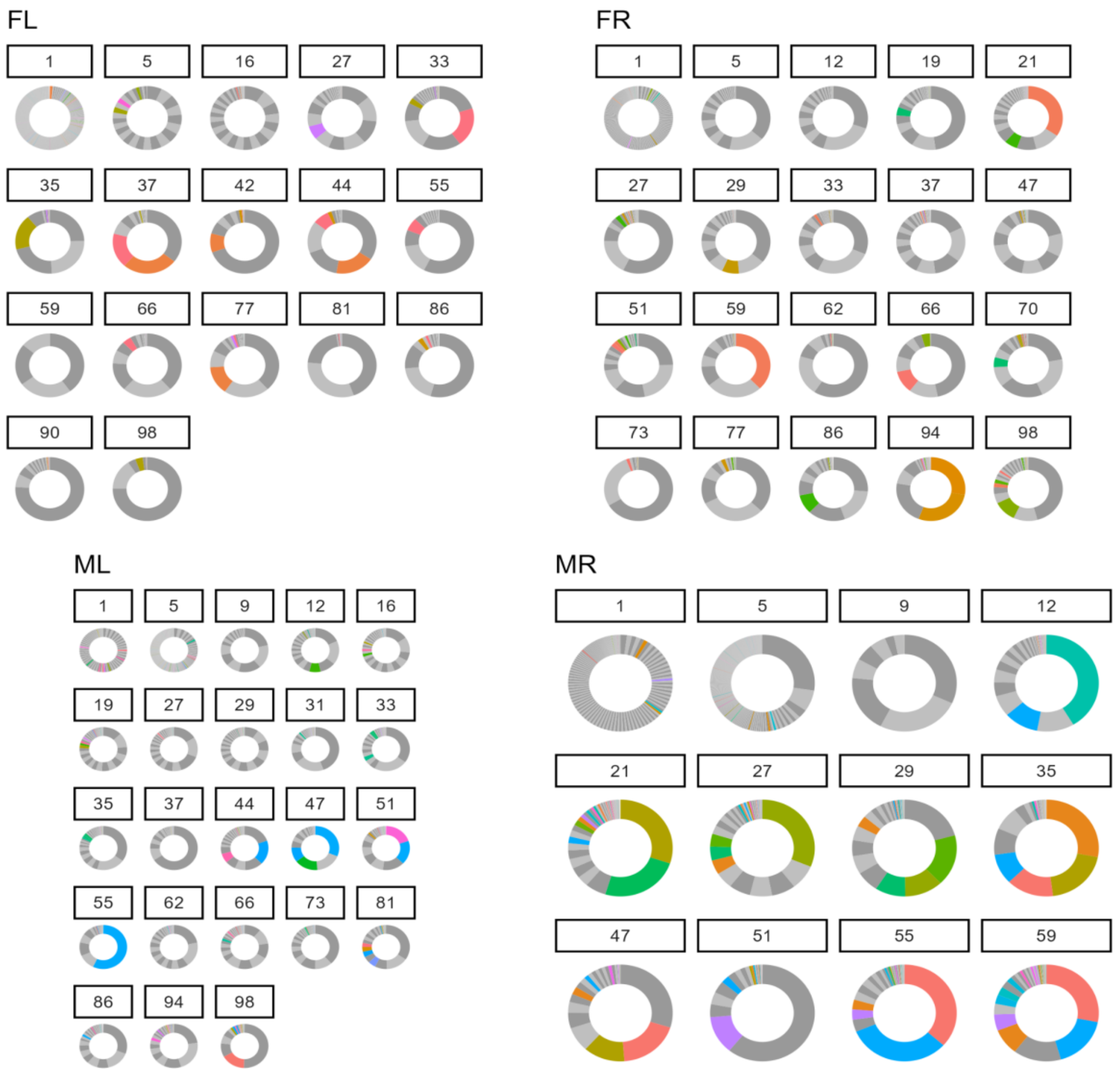


Figure 2C

Purpose:

Procedure: The longitudinal urine dataset was filtered to the top 50 kidney-enriched barcodes identified during the terminal phase of infection. For each animal and time point, the absolute levels of these selected barcodes were summed to determine the total signal magnitude specifically attributable to the kidney-derived population. These values were then visualized as longitudinal area plots (red) overlaid against the total urinary barcode load (gray), which served as a baseline reference for the entire population. This comparative mapping facilitated an assessment of absolute clonal dominance, distinguishing between the relative proportion of the population and the actual magnitude of shedding. Finally, individual animal plots were synthesized into a standardized multi-panel figure with synchronized axes for "Days Post-Injection" and "Barcode level" to enable direct cross-subject comparison of viral or cellular population dynamics.

Result: As shown in the overlay plots, urine shedding dynamics varied substantially across animals, with clear differences in both timing and magnitude of kidney-derived barcode contributions. In FR, ML, and MR, the red areas increase sharply at specific time points, indicating transient dominance of a small subset of kidney-enriched barcodes in urine, often coinciding with peaks in overall urine signal. In contrast, FL displays consistently minimal red signal across all days despite fluctuations in the gray background, indicating persistently low contribution from kidney-dominant barcodes. These patterns suggest that only certain animals undergo late-stage amplification or release of dominant kidney lineages into urine, whereas others remain in a low-shedding state. Collectively, the figure demonstrates that urine barcode profiles do not uniformly mirror total urine output but instead reflect animal-specific differences in kidney colonization, lineage expansion, and release dynamics.

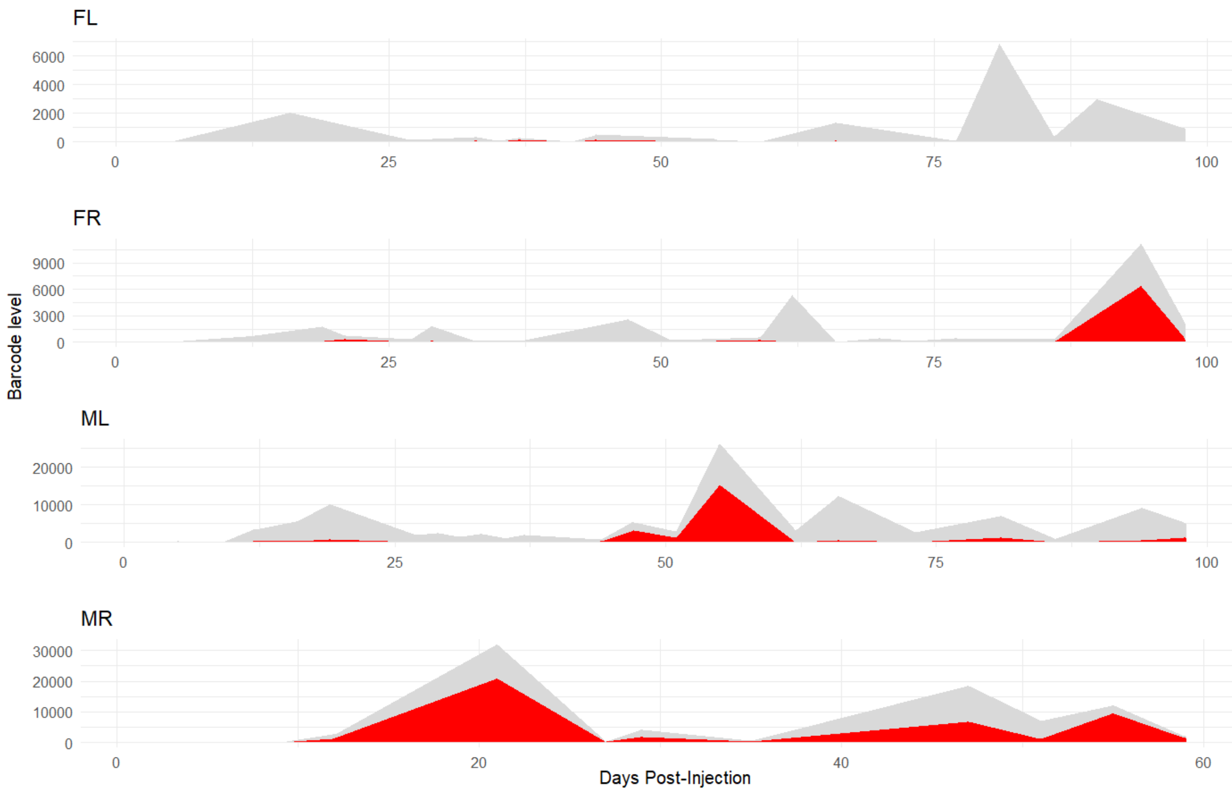


Figure 2B & 2D:

Procedure: Same as figure 2A but determine the barcodes that are absent from last 2 time points:

For each animal, barcodes detected in urine during the final two post-infection days were identified and used to define a late-stage urine exclusion set. Kidney barcode data were then filtered to retain only barcodes that were **absent from late-stage urine**. These kidney-restricted barcodes were summed across tissue samples, ranked by total abundance, and the top 50 barcodes were selected. The abundance of these kidney-enriched but urine-absent barcodes was subsequently tracked in urine across all time points using donut plots, with highlighted segments indicating the selected barcodes and background segments representing all remaining urine barcodes.

Result: The plot demonstrates that barcodes explicitly defined as absent from the final two urine time points remain largely excluded from late-stage urinary shedding across the experiment. For each animal (FL, FR, ML, MR), the donut plots represent the top 50 kidney-enriched barcodes that showed no detectable urine signal at the last two post-infection time points. Across all panels, the donuts are dominated by grey segments, indicating that these barcodes contribute minimally to the overall urine barcode pool at most time points. A small subset of barcodes exhibits weak and transient urine signals at earlier time points, but these signals are inconsistent across animals and do not persist into late infection. Together, these observations confirm that absence at the final two time points reflects a sustained lack of contribution to urinary shedding.

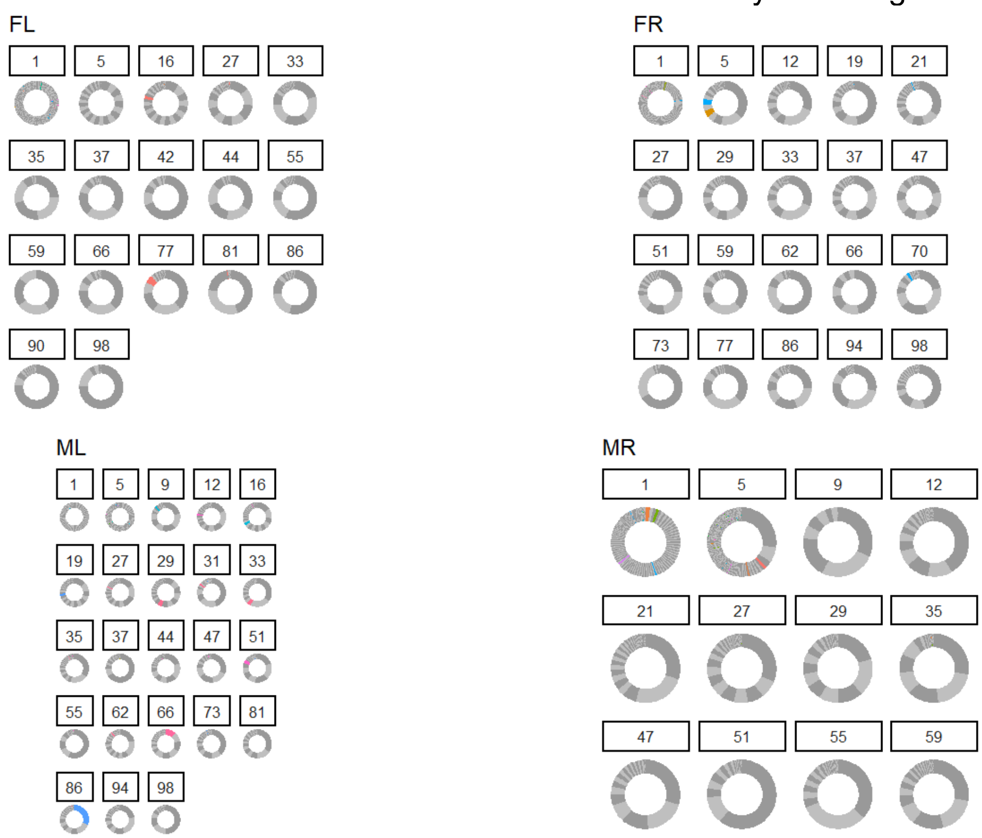


Figure 2D:

Procedure: Same as 2A, but we selected top 50 kidney-enriched barcodes absent from the final two urine time points

Result:In the analysis of kidney-ranked barcodes that were absent from urine during the final *K* collection days, only animal **ML** exhibited some detectable urine signal attributable to this barcode set. Specifically, a transient contribution was observed between approximately **days 60 and 75 post-injection**, whereas no corresponding signal was detected in **FL, FR, or MR** at any timepoint. In these latter animals, barcodes with high kidney load remained undetectable in urine throughout the study period

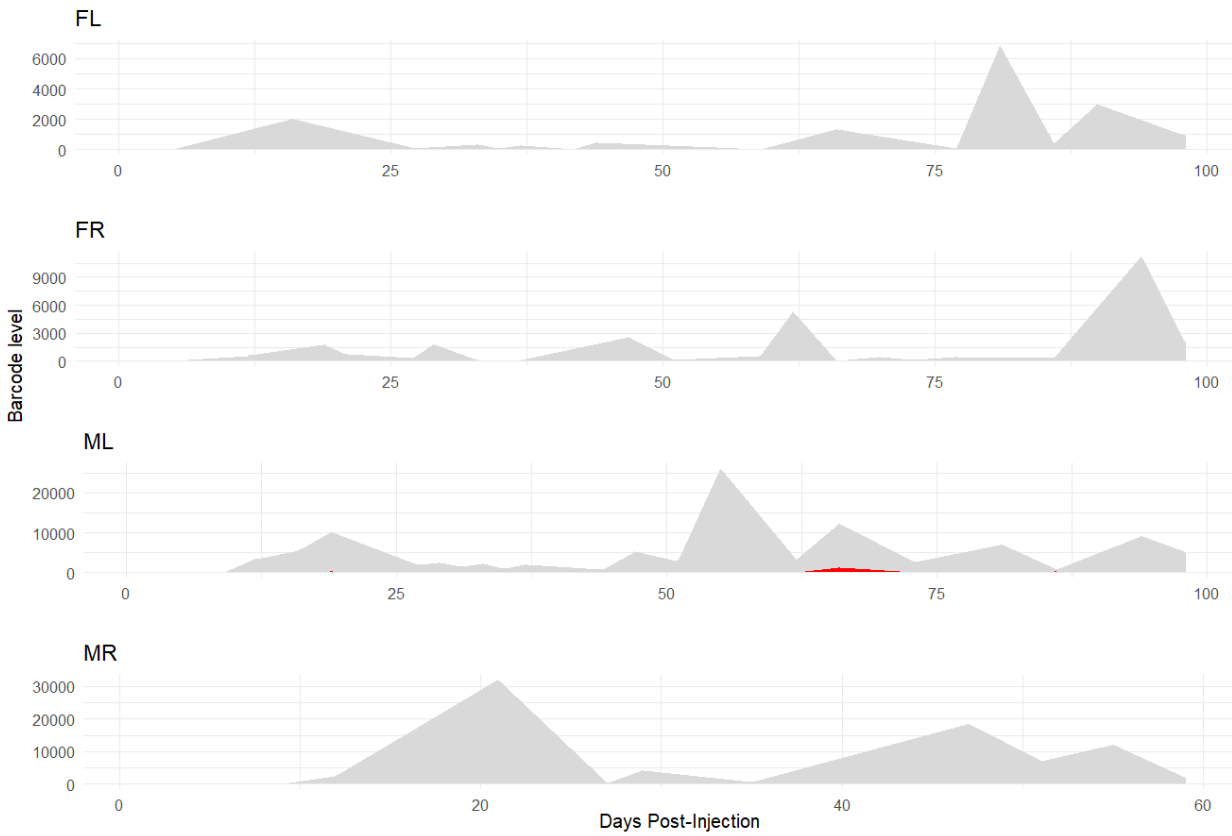


Figure 6

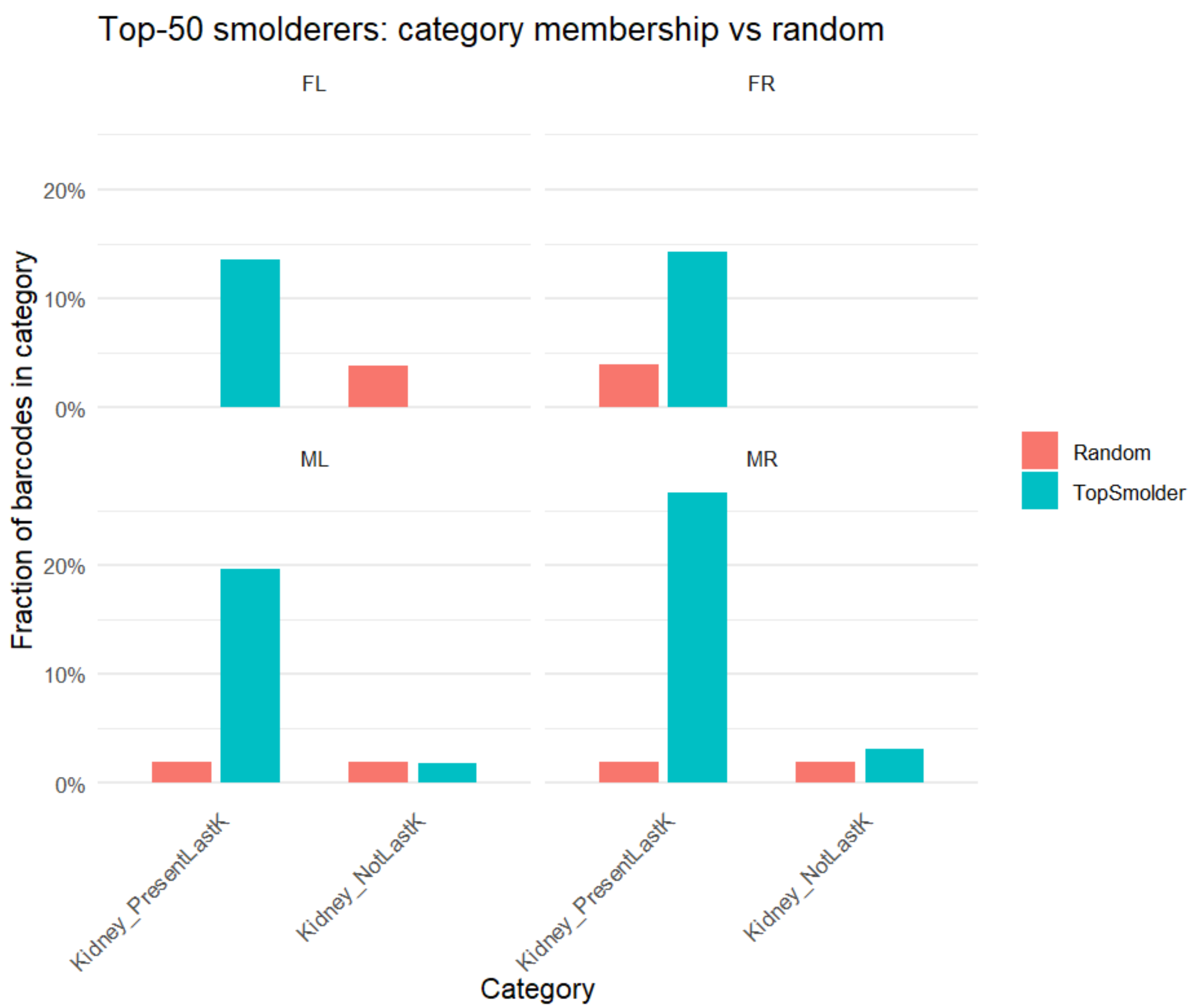
Definition: **Smolders**- defined as a barcode that is consistently detectable in urine across a large fraction of sampled time points, irrespective of its peak abundance at any single time point.

Procedure: To quantify persistent urine shedding, we defined a “smoldering” metric that captures the temporal consistency of barcode detection rather than peak abundance. For each animal and barcode, urine barcode levels were summed within each day post-injection, and a barcode was considered detected on a given day if its summed level exceeded a minimal detection threshold. The number of days on which each barcode was detected was then normalized by the total number of urine sampling days to compute the percentage of detected days. Barcodes were ranked by this percentage for each animal, and the top 50 barcodes with the highest detection frequencies were designated as smoldering lineages for downstream enrichment and comparative analyses.

For null comparision, an equal number of barcodes were randomly sampled from the urine barcode universe for each animal, excluding Top Smolderers. In parallel, barcode sets representing kidney-related and urine-related phenotypes were defined, including kidney-enriched barcodes present or absent in late urine samples, high kidney-load barcodes, and late high-shedding urine barcodes. Top Smolderers and random controls

were annotated with these category memberships, and enrichment was quantified as the fraction of barcodes falling into each category. Statistical significance of enrichment was assessed using Fisher's exact tests performed separately for each animal and category. Finally, enrichment results were visualized by comparing category membership fractions between Top Smolderers and random barcodes across animals.
Result: This figure shows that smoldering barcodes , which defined as those persistently detectable in urine across many time points, are strongly enriched for barcodes that are present in the final two urine time points (Kidney_PresentLastK) compared with randomly selected barcodes. Across all animals, a substantially larger fraction of Top Smolderers falls into the Kidney_PresentLastK category, indicating that barcodes that continue to be detected at the terminal stage of infection are also those that tend to persist at low but detectable levels throughout the time course. In contrast, barcodes classified as Kidney_NotLastK that did not detect in the final two urine samples are rare among Top Smolderers and occur at similar or only slightly elevated frequencies relative to random barcodes. Together, these results demonstrate that urine persistence ("smoldering") is not a stochastic property but is tightly linked to sustained detectability into late time points, consistent with ongoing or continuous kidney-derived shedding rather than transient early release

In ML and MR, smoldering barcodes are absent from the Kidney_NotLastK category, meaning that barcodes do not show persistent, low-level detectability across the time course. In other words, in these animals, smoldering and late detectability are tightly coupled: a barcode either persists into the last K days and smolders throughout, or it drops out entirely and does not exhibit long-term urine presence. This contrasts with FL and FR, where a small fraction of smoldering barcodes fall into Kidney_NotLastK, suggesting weaker coupling between persistence and late-stage shedding. It implies that in ML and MR, smoldering reflects a stable, sustained kidney-to-urine release state, rather than intermittent or transient shedding



Top-50 smolderers: category membership vs random
FL
FR
ML
MR
20%
10%
0%
Fraction of barcodes in category
Random
TopSmolder
Kidney_PresentLastK
Kidney_NotLastK
Category

Additional Analysis

<u>Definition</u>:

Smolderer: Top-50 barcodes persistently detected in urine across time.
Kidney_PresentLastK: Kidney-enriched barcodes detectable in the final 2 timepoints' urine samples.
Kidney_HighLoad: Top-50 barcodes by total kidney abundance.
Urine_LateHighShed: Top-50 barcodes with the highest urine abundance during final 2 timepoints' urine samples.

Procedure:

To compare overlap among barcode populations, barcode identities were treated as animal-specific, such that the same barcode number detected in different animals was considered a separate lineage. For each animal, the top 50 barcodes were selected for four categories: persistently detected ("smoldering") urine barcodes, kidney-associated barcodes detected in the final two urine time points, kidney barcodes with the highest overall tissue abundance, and barcodes exhibiting the highest urine shedding during the terminal period. These barcode sets were compared using an UpSet analysis to quantify the size of each group and the extent of overlap among smoldering, kidney-associated, and late urine–shedding barcode populations.

Result: The plot shows that most barcodes are category-specific, with limited overlap among smoldering, kidney-associated, and late urine–shedding groups. Overlap is most evident between smoldering barcodes and those detected in the final two urine time points, whereas intersections with high kidney load or late high urine shedding alone are comparatively sparse. Only 13 barcodes are shared across all four categories, indicating that lineages simultaneously exhibiting persistence, high kidney burden, and late urine shedding are rare. Overall, these results suggest that barcode persistence reflects a distinct behavior rather than a general consequence of high tissue abundance or late-stage shedding.

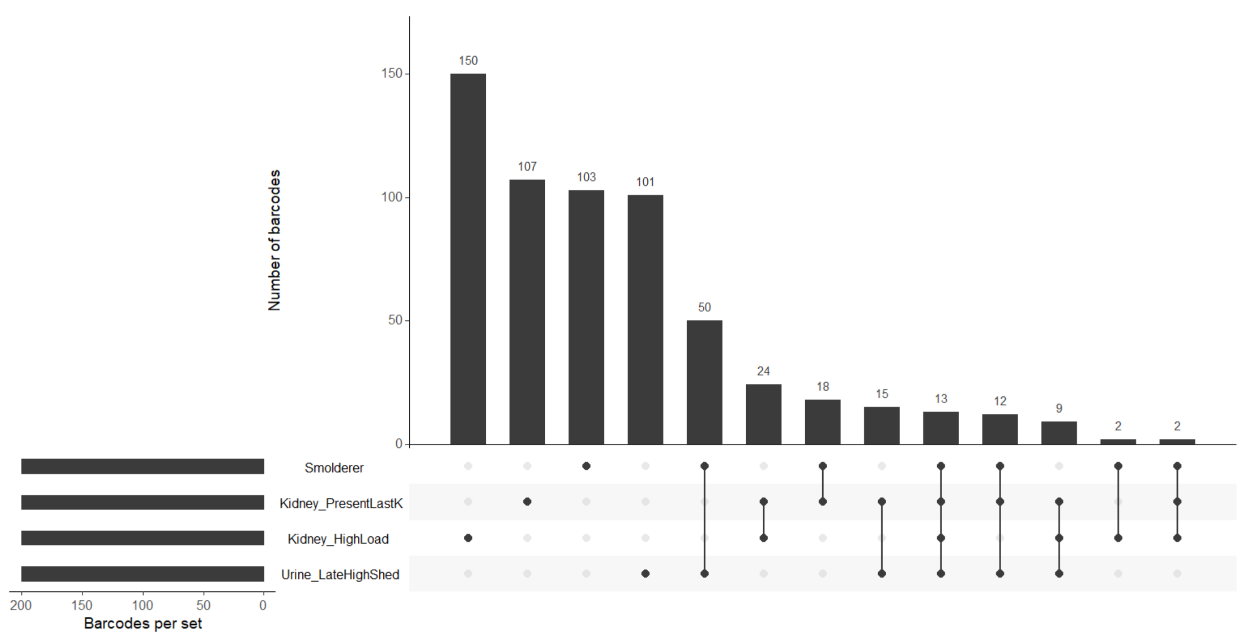
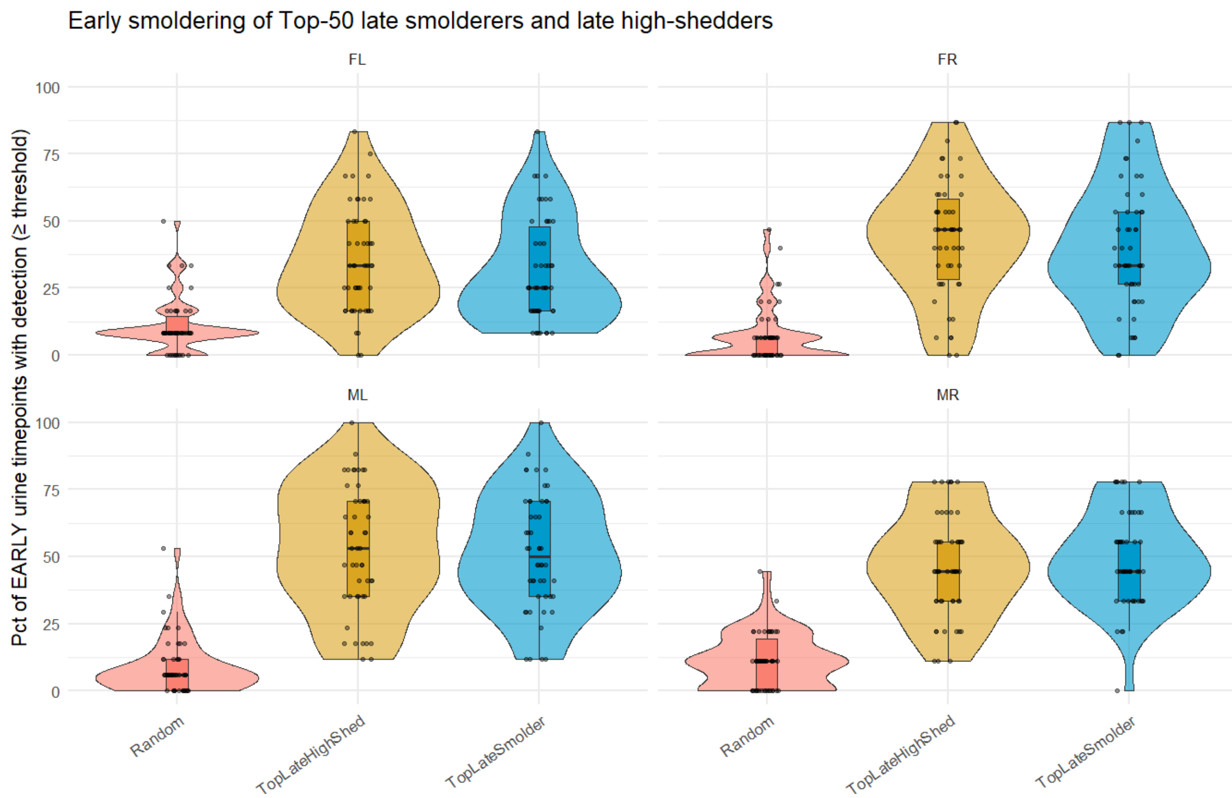


Figure 7

Procedure: Urine sampling days within each animal were divided into early and late periods by ordering time points by days post-injection and defining the late period as the final quartile of available time points. Urine barcode levels were then aggregated for each animal, barcode, and day by summing barcode abundance within each time point, and detection was defined as a summed daily barcode level exceeding a minimal threshold. For each barcode, the number and percentage of early and late time points (late defined as the final 25% of timepoints, and early defined as first 75%) with detectable signals were calculated, along with the total barcode load across late time points. Using these metrics, three barcode sets were defined per animal: (i) TopLateSmolder, the top 50 barcodes ranked by the frequency of detection during late time points (tie-broken by late total load); (ii) TopLateHighShed, the top 50 barcodes ranked by total late-period load; and (iii) a Random set of 50 barcodes sampled from the urine barcode universe after excluding barcodes in the two top-ranked sets. All selected barcodes were annotated with their early detection percentages. Finally, the distribution of early detection percentages was compared across the three groups using animal-faceted violin/box/jitter plots, enabling evaluation of whether barcodes that dominate late-period persistence or shedding also exhibit consistent detectability during the early phase.

Result: It shows that barcodes classified as **late smolderers** or **late high-shedders** also tend to be detectable during the **early phase** of the experiment, whereas randomly selected barcodes rarely show early detection. Across all animals, both TopLateSmolder and TopLateHighShed groups display substantially higher percentages of early urine time points with detectable signal compared with the random control, indicating that barcodes contributing to late persistence or late shedding are typically not newly emerging at late stages. Instead, these barcodes are already present at low to moderate levels earlier in the time course. In most animals, the early detection distributions of late smolderers and late high-shedders overlap considerably, suggesting that persistence into late stages and high late-stage contribution both arise from lineages that establish early. Together, these results support a model in which late-dominant urine barcodes reflect sustained lineage presence rather than transient late-stage expansion.



[ALT]: Same plot but with different random subset- yields the **same qualitative conclusion**

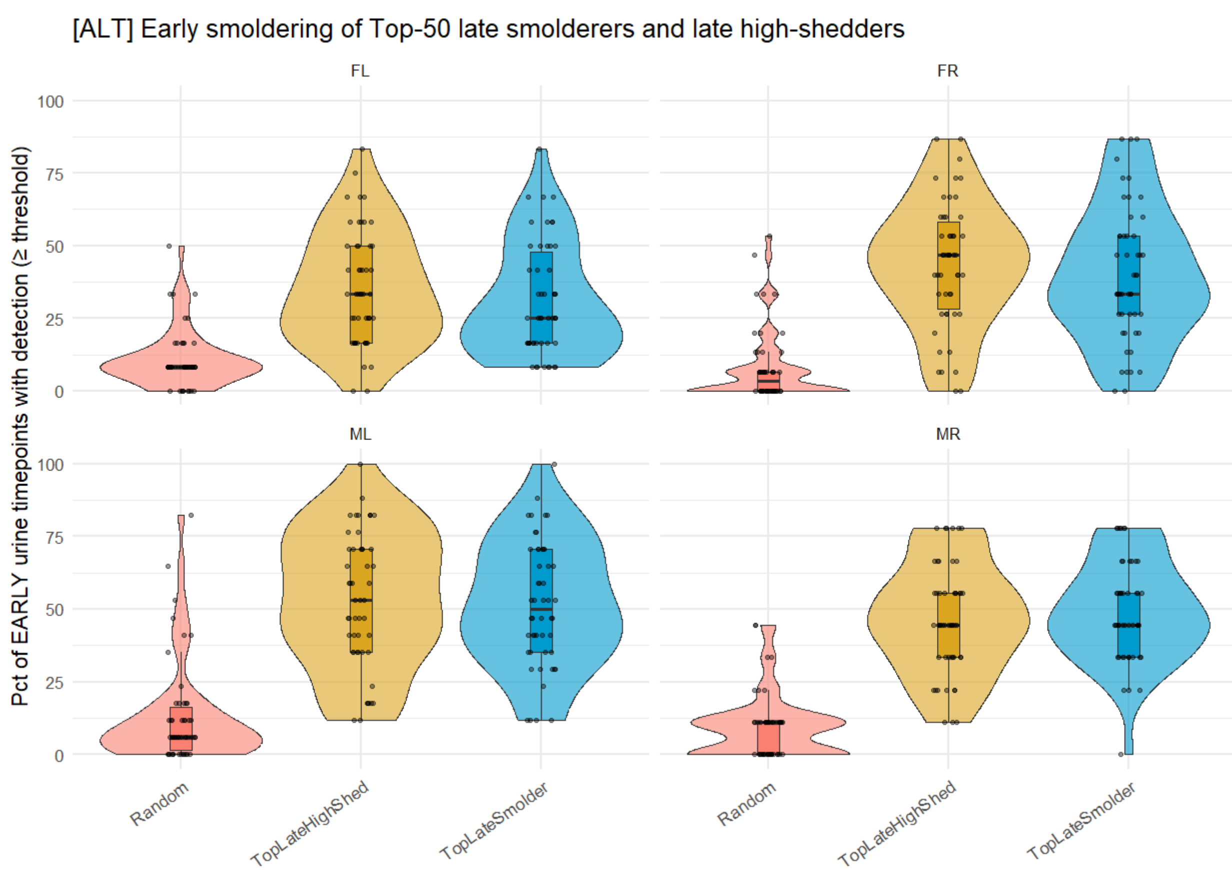


Figure 8 and S1:

Procedure: Kidney-associated barcodes were stratified based on their presence or absence in late-stage urine to evaluate tissue-specific barcode distribution patterns. Barcode abundance data were first restricted to kidney tissue samples. Kidney barcodes detected in urine during the final two post-infection time points were classified as kidney_presentLastK, whereas kidney barcodes absent from late-stage urine were classified as kidney_notLastK. For each animal, barcode lists for both groups were compiled without applying an abundance threshold, ensuring inclusion of all qualifying kidney barcodes.

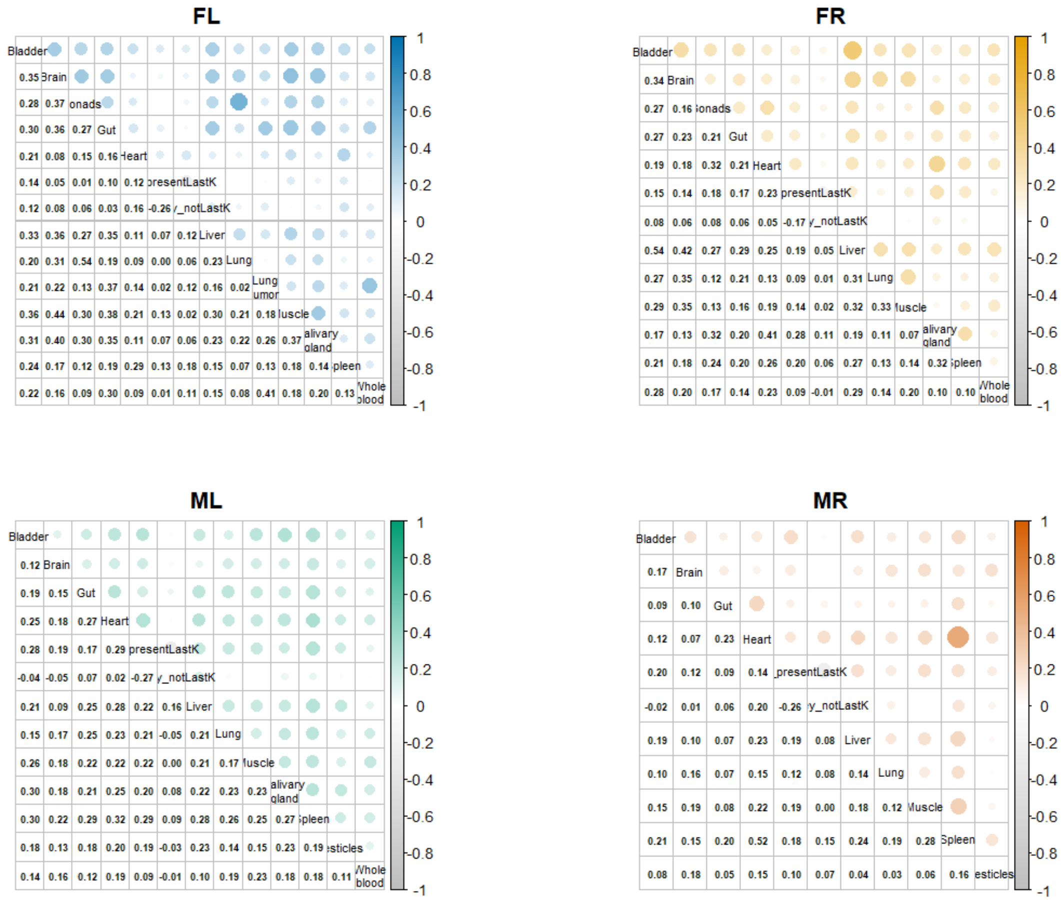
For within-animal comparisons, barcode abundance data were subset to individual animals and restricted to barcodes belonging to either kidney group. Kidney observations were relabeled into two distinct compartments (kidney_presentLastK and kidney_notLastK), while all other organs retained their original labels. Barcode abundance values (fractional weighted counts) were reshaped into a barcode-by-organ matrix with missing values filled as zero. Pairwise Spearman correlation coefficients were then calculated across organ compartments to quantify similarity in barcode abundance patterns within each animal.

For between-animal comparisons, kidney barcode abundance profiles were analyzed separately for the kidney_presentLastK and kidney_notLastK groups. Barcode abundance values were reshaped into barcode-by-animal matrices, and pairwise Spearman correlations were computed between animals for each group. These correlations were visualized to assess the consistency of kidney barcode distributions across mice for barcodes that either contributed to late urinary shedding or remained absent from urine.

All correlation analyses used Spearman's rank correlation to account for non-normal barcode abundance distributions and differences in scale.

Result: Across all four animals (FL, FR, ML, MR), the Spearman correlation matrices show predominantly weak-to-moderate positive associations among tissues, with most correlation coefficients remaining very low (generally ~0.10–0.40), indicating only limited coordination of barcode abundance patterns across organs. Importantly, the late-urine detection label (**presentLastK**) does not exhibit strong or consistent correlation with any single tissue compartment, suggesting that barcode persistence in the final urine timepoints is not clearly explained by enrichment in a particular organ based on this correlation structure alone. The kidney barcode group absent from late urine (**y_notLastK**) also shows largely near-zero correlations with tissues and is modestly negatively correlated with **presentLastK**, which is expected given that these two categories represent opposing barcode definitions. Among the animals, **ML** displays the most internally consistent pattern of positive correlations, aligning with the observation that ML is the only animal showing a clearer detectable urine signal at later timepoints

(60–75); however, overall, these results suggest substantial variability and limited evidence for a robust, reproducible tissue-driven signature distinguishing late-urine–present versus late-urine–absent kidney barcodes.



Result: The Spearman correlation analysis shows a contrast between kidney barcodes that are absent from late urine (NOT last-K) and present in late urine (last-K). For the kidney_absent set, correlations between animals are consistently moderate and negative (ρ≈ −0.27 to −0.33). It indicates that the relative ranking of these barcodes in the kidney is not conserved across mice. This anti-correlation suggests strong animal-specific effects and poor cross-animal reproducibility for kidney barcodes that fail to appear in late urine. In contrast, the kidney_present (last-K) set shows correlations that are near zero across all animal pairs (|ρ| ≤ 0.05). This indicates an absence of shared rank structure rather than systematic agreement or disagreement, consistent with stochastic or idiosyncratic late-urine appearance across animals.
These results demonstrate that neither kidney barcodes absent from late urine nor those present in late urine exhibit a conserved inter-animal pattern. The negative correlations in the absent set further emphasize heterogeneity, while the near zero correlations in the present set indicate a lack of reproducible late-shedding signatures.

Overall, late urine barcode behavior appears dominated by animal-specific variability rather than a common kidney barcode.